Med Assist

An Assistive Mental Health Platform

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Advised by: Dr. Dennis Wu

Repo

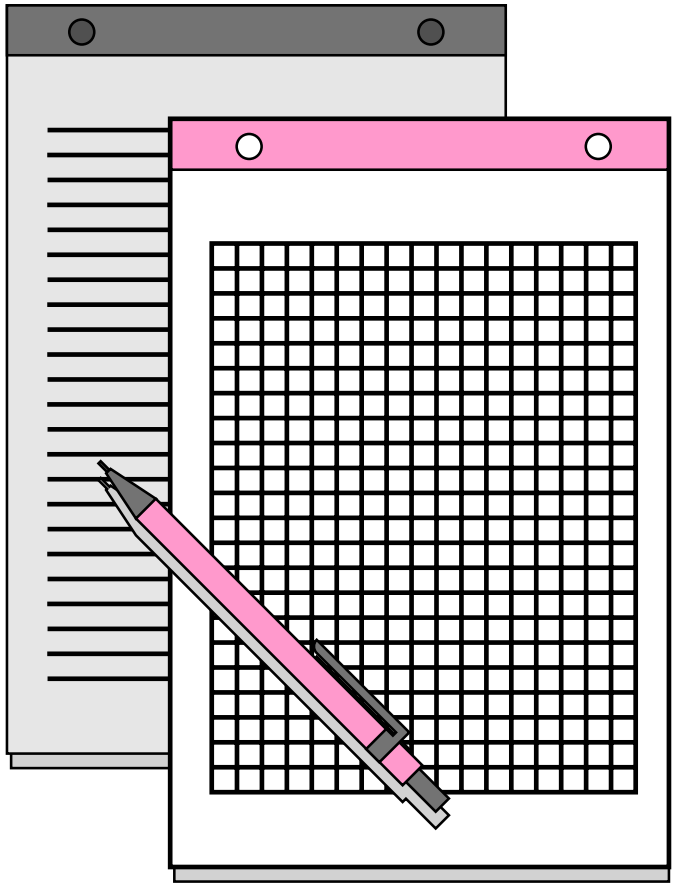
Purpose and Goal Statements

Purpose:

Offer accessible and caring mental-health conversations when professional help may not be available or affordable.

Goal Statements:

- Integrate LLM with RAG to give accurate and relevant responses
- Enable users to track mental health trends and receive recommendations
- Keep user data private and secure while storing helpful summaries for later review
- Offer professional alternatives depending on user needs



Project Abstract

MedAssist is an assistive mental health platform powered by a large language model (LLM) with retrieval-augmented generation (RAG). It offers a free, accessible, and unbiased alternative to mental health care. The system connects an LLM to trusted DSM-5 knowledge to give accurate, context-aware responses. Through an interactive chat interface, users are able to express their thoughts, monitor mood patterns, and track personal growth over time. Alternative resources are also provided for users requiring more professional medically-focused assistance.

DSM-5: Diagnostic and Statistical Manual of Mental Disorders, 5th Edition

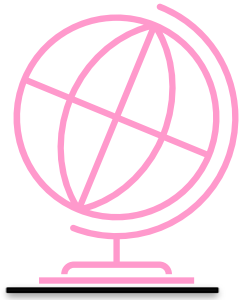
Intellectual Mertis

MedAssist integrates LLM with RAG grounded in DSM5 knowledge to reduce hallucination and improve contextual relevance.

Beyond basic question answering, the design supports extensions such as mood trend analysis, journaling insights, and personal health dashboards.

The project includes quantitative evaluation (hallucination rate, response correctness, faithfulness, and relevancy), which supports future research and improvement of the assistant.

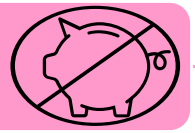
Broader Impacts



Private, judgment-free support:

- MedAssist gives users a 100% free, secure, and confidential space to explore their mental health. Featuring a chatbot that allows users a less stressful environment to reflect and learn without the pressure of opening up to another person.

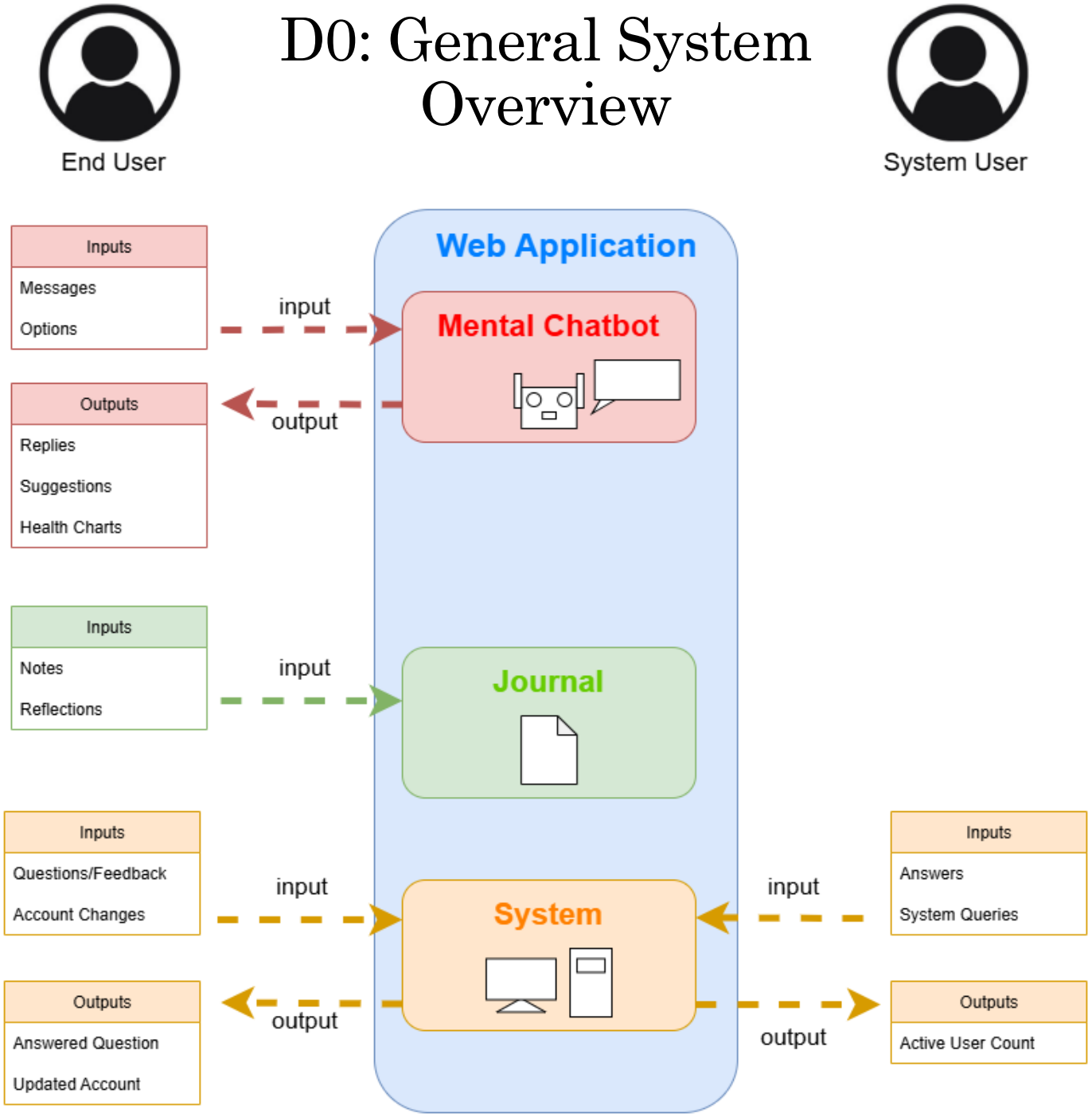
Cost-free and accessible:



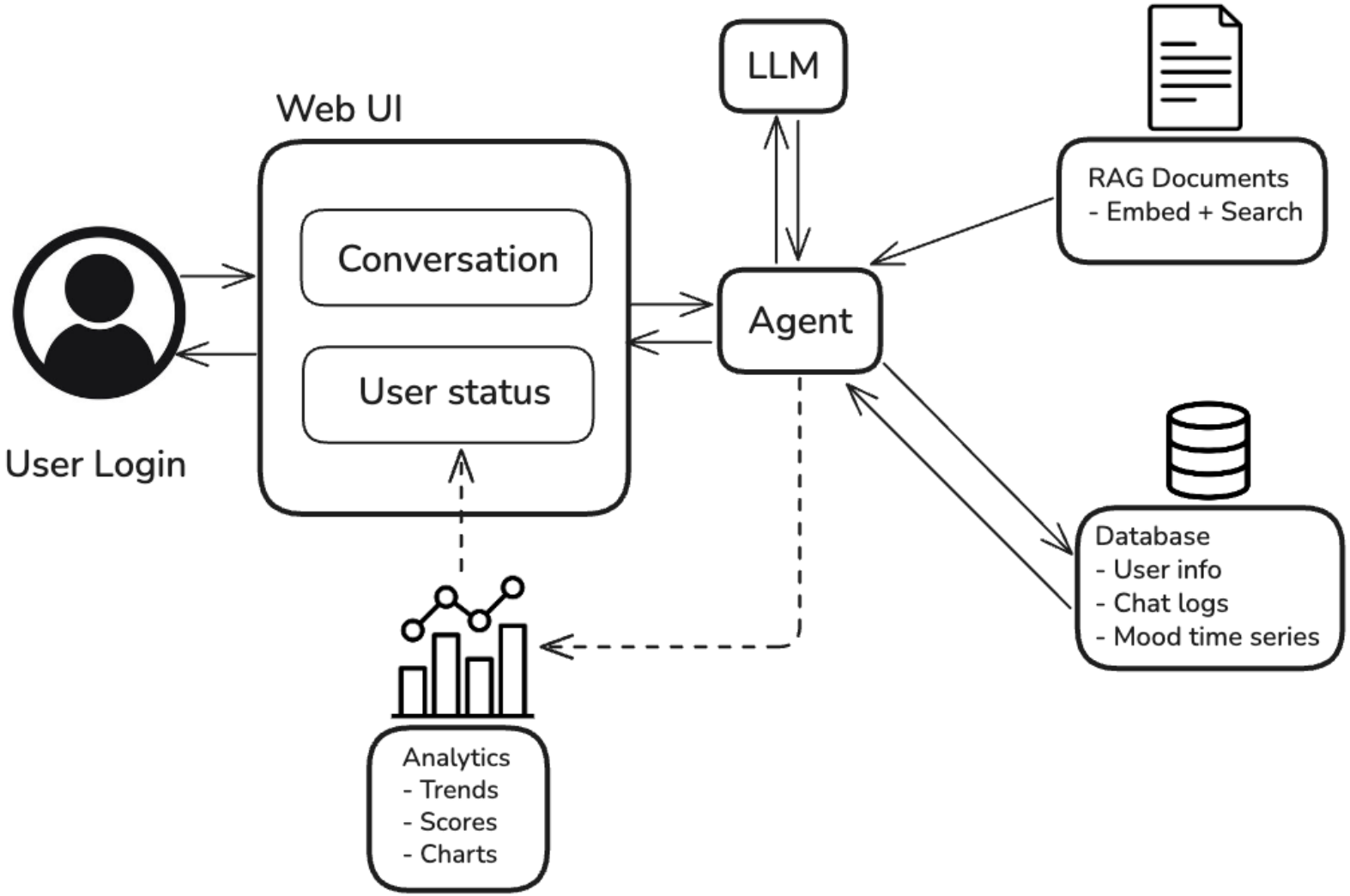
- Traditional mental-health services can be expensive; this often discourages people from seeking help. MedAssist helps remove this barrier by offering helpful tools at no-cost for users to learn more and improve their own mental health.

Design Specificat

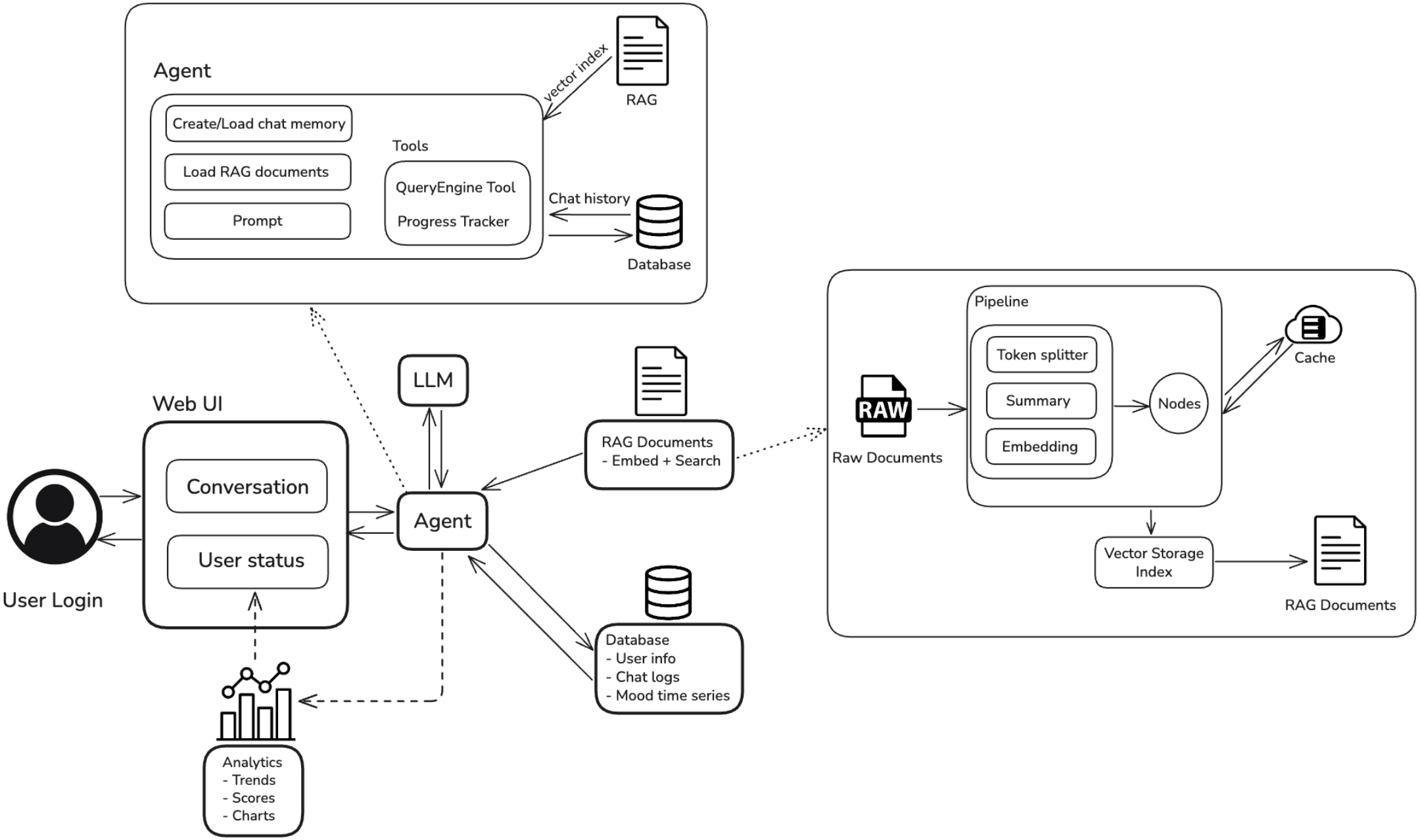
D0: General System Overview



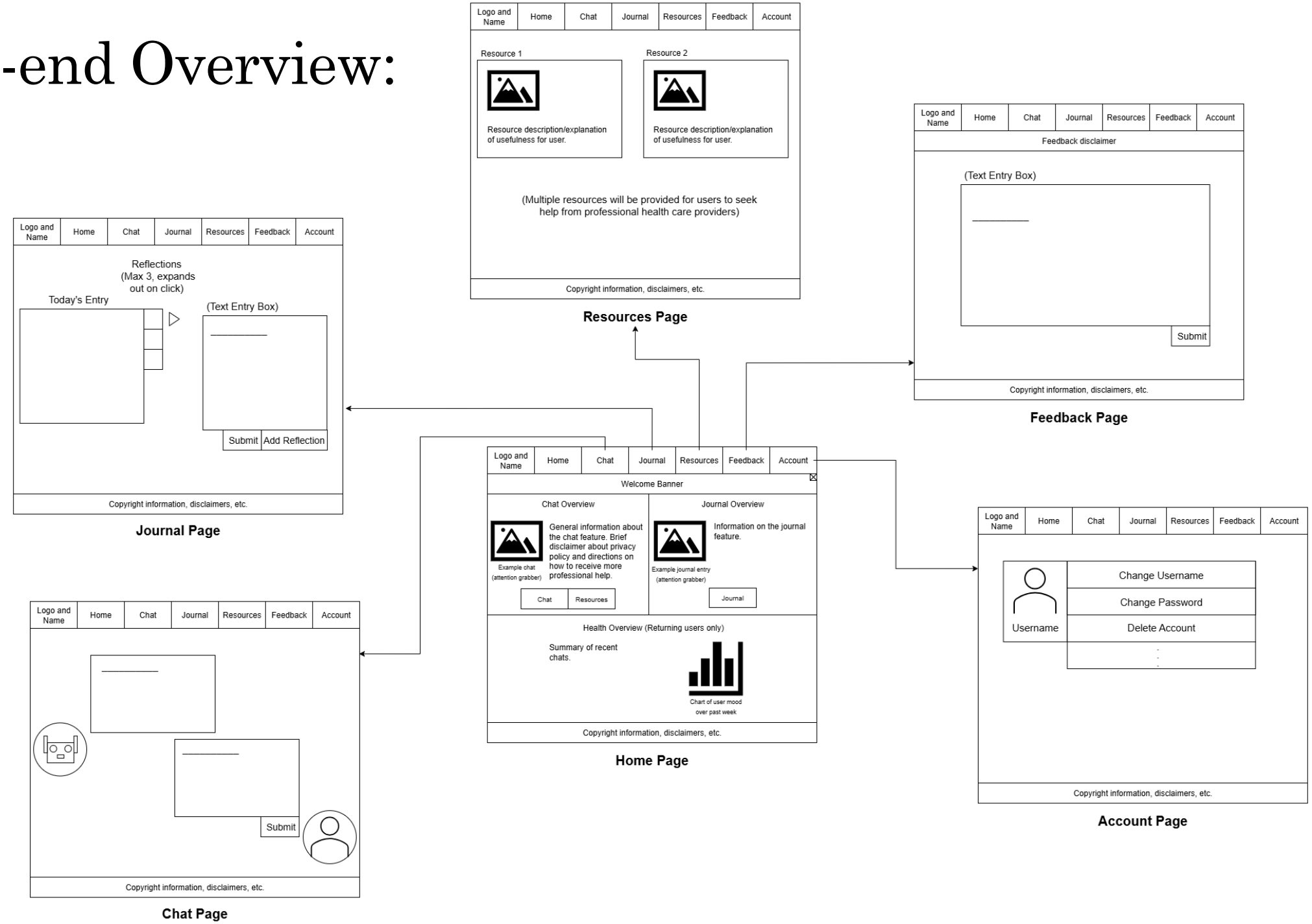
D1: Chatbot System Overview



D2: Advanced Chatbot Overview



Front-end Overview:



Technologies

Frontend

- HTML + CSS
- JavaScript
- React
- Streamlit (Fast demo)

Backend

- Python
- FastAPI
- LlamaIndex (RAG, LLM)

Database

- Vector DB (Knowledge retrieval)
- PostgreSQL (User & Application data)

Milestones

Milestone	Focus	Deliverables	Date
1. Project Initialization	Define scope & team expectations	Team contract, proposal, requirements, initial timeline	August 2025
2. Research	Build foundational knowledge	DSM-5 sources, resource list, legal/ethical constraints	September 2025
3. System Design	Plan system architecture & user experience	UI/UX wireframes, database schema, RAG pipeline design	October – November 2025
4. Protype Creation	Early functional model	Front-end mock, back-end skeleton, & API diagram	December 2025
5. Implementation	Build entire system	Integrated front/back end, functional RAG pipeline & chat generation	February – March 2026
6. Testing	Validate & ensure functionality	Unit tests, mock tests, bug fixes	March 2026
7. Final Integration & Demo	Polish and finalize	Final build, expo presentation/materials	April 2026

Current Results

You are here

Frontend:

- UI/UX
- Journal, Feedback System

Backend:

- Database
- RAG Pipeline
- API

Implementation

- Load DSM-5 content
- Integrate front/back end

Testing

- Unit/Mock tests
- Bug fixes

Finalize Integration

- Polish UI
- Update documentation

In order to finish our project, we will need to complete the implementation phase. Once the front/back end have been paired and the DSM-5 content loaded into the model, we will be able to perform testing and make any minute changes required to finalize the project.

Challenges

Challenge	Scope	Solution
Limited GPU access and paid API costs	Economic	Use of lightweight models and optimized RAG queries
Chatbot provides support only (not diagnosis/treatment)	Ethical	Clearly defined scope, addition of guardrails, redirection of users to professional resources
Must protect user privacy	Ethical	Minimal data collection and enforcement of strict access controls
Need to encrypt data and protect API keys	Security	Made use of environment variables & secret managers + end-to-end data encryption
Must follow API and open-source licensing	Legal	Reviewed licenses, documented all decencies, and ensured attribution & compliance
Need disclaimers to avoid misrepresentation	Legal	Displayed disclaimers in UI, added periodic reminders including links to verified resources
Use simple, non-judgmental, jargon-free language	Social & Cultural	Applied plain-language guidelines, empathetic phrasing, and tested clarity on a range of diverse users

Thank you!

Repository: https://github.com/toan-ly/Senior-Design